Kevin D. Stubbs

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Research Interests

My research develops and analyzes computational methods for quantum many-body systems, with a focus on electronic structure in real-world materials. My recent work centers on models of interacting electrons in twisted bilayer graphene and other moiré materials. Previously, I have worked on constructing localized Wannier functions in both periodic and non-periodic materials for fast, sparse calculations. I also work on developing mathematical approaches to rigorously analyze these models and methods.

Topics: Computational mathematics, electronic structure theory, computational materials science, Hartree–Fock and quantum embedding methods, Wannier functions, moiré materials.

Positions

Interdisciplinary Postdoctoral Associate 2026 – Present
School of Mathematics, University of Minnesota – Twin Cities
Mentors: Mitchell Luskin, Alexander Watson

Postdoctoral Researcher 2022 – 2026
Department of Mathematics, University of California, Berkeley
Mentor: Lin Lin
Supported by the Simons Foundation Targeted Grant "Moiré Materials Magic". Research only position.

Simons Postdoctoral Fellow 2021 – 2022
Institute for Pure and Applied Mathematics, University of California, Los Angeles
Mentor: Stanley Osher
Research only position.

Education

Duke University 2015 – 2021
Ph.D. Mathematics

- *Advisor:* Jianfeng Lu
- *Thesis Title:* "On Exponentially Localized Wannier Functions in Non-Periodic Insulators"

University of Maryland – College Park 2011 – 2015
B.S., Mathematics (High Honors) and B.S., Computer Engineering (cum laude)

Fellowships and Awards

- Simons Postdoctoral Fellowship (2021-2022).
- National Science Foundation Graduate Research Fellowship (2017-2020).
- Advanced Short-Term Research Opportunity at Oak Ridge National Lab (May 2019).
- February Fourier Talks Travel Award (February 2018).
- Selected Participant 8th Gene Golub SIAM Summer School (2017).

Papers and Manuscripts

Papers in physics journals follow the physics convention: authors are ordered by contribution, with senior author(s) last. * denotes a student coauthor under my supervision. † denotes co-first authors.

Preprints

[21] S. Escobar, L. Lin, M. Ragone, and **K. D. Stubbs**. "On the Metastability of the Mean-Field Interchange Model for Local Dimension ≥ 3," *arXiv preprint*, 2026, arXiv:2610.01987, submitted to *QIP 2027*.

[20] T. Kong*, A. B. Watson, L. Lin, M. Luskin, and **K. D. Stubbs**. "Relaxation Effects on Hartree–Fock Ground States in Twisted Bilayer Graphene at Even Integer Fillings," *arXiv preprint*, 2026, arXiv:2609.02815, submitted to *Physical Review B*.

[19] Z. Huang and **K. D. Stubbs**. "Exactly Flat Bands beyond the Chiral Limit in Bistritzer–MacDonald Hamiltonians," *arXiv preprint*, 2026, arXiv:2608.31147, submitted to *Probability and Mathematical Physics*.

[18] **K. D. Stubbs** and M. Zworski. "Hartree–Fock Coercivity for Twisted Bilayer Graphene," *arXiv preprint*, 2026, arXiv:2608.23884, submitted to *Communications in Mathematical Physics*.

[17] J. Basso, T. Bergamaschi, L. Lin, M. Ragone, and **K. D. Stubbs**. "Spectral Gap of the Davies Generator for the Mean-Field Heisenberg Model," *arXiv preprint*, 2026, arXiv:2607.21798, submitted to *Communications in Mathematical Physics*.

Published Papers

[16] J. Lu and **K. D. Stubbs**. "Algebraic localization implies exponential localization in non-periodic insulators," *Archive for Rational Mechanics and Analysis*, 2026, 10.1007/s00205-026-02242-z.

[15] W. Kim†, R. Kim†, **K. D. Stubbs**†, S. Louie, and L. Lin. "Ab initio quantum embedding description of magic angle twisted bilayer graphene at even-integer fillings," *Physical Review B (Editor's Suggestion)*, 2026, 10.1103/mp1d-6nnl.

[14] **K. D. Stubbs**, M. Ragone, A. H. MacDonald, and L. Lin. "The Many-Body Ground State Manifold of Flat Band Interacting Hamiltonian for Magic Angle Twisted Bilayer Graphene," *Communications in Mathematical Physics*, 2026, 10.1007/s00220-026-05736-9.

[13] **K. D. Stubbs**, S. Becker, and L. Lin. "On the Hartree–Fock Ground State Manifold in Magic Angle Twisted Graphene Systems," *Multiscale Modeling & Simulation*, 2025, 10.1137/24M1689909.

[12] T. Kong*, A. B. Watson, M. Luskin, and **K. D. Stubbs**. "Interacting Twisted Bilayer Graphene with Systematic Modeling of Structural Relaxation," *Electronic Structure*, 2025, 10.1088/2516-1075/adeb25.

[11] S. Becker, L. Lin, and **K. D. Stubbs**. "Exact Ground State of Interacting Electrons in Magic Angle Graphene," *Communications in Mathematical Physics*, 2025, 10.1007/s00220-025-05300-x.

[10] A. Alase, **K. D. Stubbs**, B. C. Sanders, and D. L. Feder. "Erasure Conversion in Majorana Qubits via Local Quasiparticle Detection," *Physical Review Research*, 2024, 10.1103/PhysRevResearch.6.043294.

[9] J. Lu and **K. D. Stubbs**. "Algebraic localization of Wannier functions implies Chern triviality in non-periodic insulators," *Annales Henri Poincaré*, 2024, 10.1007/s00023-024-01444-z.

[8] F. M. Faulstich†, **K. D. Stubbs**†, Q. Zhu†, T. Soejima, R. Dilip, H. Zhai, R. Kim, M. P. Zaletel, G. K.-L. Chan, and L. Lin. "Interacting models for twisted bilayer graphene: A quantum chemistry approach," *Physical Review B (Editor's Suggestion)*, 2023, 10.1103/PhysRevB.107.235123.

[7] J. Lu, **K. D. Stubbs**, and A. B. Watson. "Existence and computation of generalized Wannier functions for non-periodic systems in two dimensions and higher," *Archive for Rational Mechanics and Analysis*, 2022, 10.1007/s00205-021-01721-9.

[6] S. Brandsen, M. Lian, **K. D. Stubbs**, N. Rengaswamy, and H. D. Pfister. "Adaptive procedures for discriminating between arbitrary tensor-product quantum states," *Physical Review A*, 2022, 10.1103/PhysRevA.106.012408.

[5] S. Brandsen, **K. D. Stubbs**, and H. D. Pfister. "Reinforcement Learning with Neural Networks for Quantum Multiple Hypothesis Testing," *Quantum*, 2022, 10.22331/q-2022-01-26-633.

[4] **K. D. Stubbs**, A. B. Watson, and J. Lu. "Iterated projected position algorithm for constructing exponentially localized generalized Wannier functions for periodic and nonperiodic insulators in two dimensions and higher," *Physical Review B*, 2021, 10.1103/PhysRevB.103.075125.

[3] A. N. Morozovska, E. A. Eliseev, **K. D. Stubbs**, R. Vasudevan, Y. Kim, and S. V. Kalinin. "Phase diagrams of single-layer two-dimensional transition metal dichalcogenides: Landau theory," *Physical Review B*, 2020, 10.1103/PhysRevB.101.195424.

[2] W. Czaja, B. Manning, J. M. Murphy, and **K. D. Stubbs**. "Discrete directional Gabor frames," *Applied and Computational Harmonic Analysis*, 2018, 10.1016/j.acha.2016.07.004.

Book Contributions

[1] **K. D. Stubbs**. "The formalism of second quantization," Appendix A in M. Zworski and Z. Tao, *PDE Methods in Condensed Matter Physics*, Publisher: American Mathematical Society, to appear 2027, current draft, math.berkeley.edu/~zworski/Notes_279.pdf.

Invited Talks

- SIAM Mathematical Aspects of Materials Science 2027 (*upcoming*); Seattle, WA (May 2027).
- SIAM Computational Science and Engineering 2027 (*upcoming*); Pittsburgh, PA (February 2027).
- PDE Seminar; University of Minnesota – Twin Cities, Minneapolis, MN (September 2026).
- Mathematical Aspects of 2D Quantum Materials and Meta-Materials; IMSI, Chicago, IL (June 2026).
- SIAM Annual Meeting; Montréal, Canada (August 2025).
- Theoretical Chemistry Seminar; University of California, Berkeley, Berkeley, CA (April 2025).
- Mathematics Department Colloquium; University of Minnesota – Twin Cities, Minneapolis, MN (March 2025).
- International Workshop on 2D and Moiré Materials; Roscoff, France (July 2024).
- Mathematical Models of Electronic Transport & Phases in Low-Dimensional Materials; Brin Center, University of Maryland – College Park (March 2024).
- SOLID Math 2022; Trieste, Italy (September 2022).
- SOLID Math 2021; Online (August 2021).
- SIAM Mathematical Aspects of Materials Science 2021; Online (May 2021).

- UC Berkeley Quantum Many-Body Seminar; Online (January 2021).
- Oak Ridge National Laboratory Seminar; Oak Ridge, TN (August 2019).
- REU Alumni Talk; Boise State University, Boise, ID (July 2017).

Contributed Talks and Posters

- Moiré Materials Workshop (Poster Session); Flatiron Institute, New York, NY (May 2026).
- APS Global Summit 2026; Denver, CO (March 2026).
- CIQC Annual Meeting (Poster Session); Pacific Grove, CA (July 2025).
- Moiré Materials Workshop (Poster Session); Flatiron Institute, New York, NY (May 2025).
- APS March Meeting 2024; Minneapolis, MN (March 2024).
- Advancing Quantum Mechanics with Mathematics and Statistics Reunion Conference; Lake Arrowhead, CA (December 2023).
- SciDAC Principal Investigator's Meeting (Poster Session); Rockville, MD (September 2023).
- APS March Meeting 2023; Las Vegas, NV (March 2023).

Teaching and Mentoring

As Instructor of Record

- *MATH 5652 (Introduction to Stochastic Processes, University of Minnesota – Twin Cities):* Fall 2026, Spring 2027 (*scheduled*)
- *MATH 112 (Calculus I, Duke University):* Spring 2020
- *MATH 122 (Calculus II, Duke University):* Fall 2016

As Teaching Assistant

- *ECE 587 (Information Theory, Duke University):* Fall 2018
- *MATH 122 (Calculus II, Duke University):* Fall 2015

Students Supervised

- Sanjay Krishnan (High School Student; Summer 2026 – present)
  - Supervising a project on avoiding bottlenecks in mixed-spin Ising models.
- Tianyu Kong (Graduate Student from University of Minnesota – Twin Cities; 2024 – 2026)
  - Co-supervised project on modeling many-body effects in twisted bilayer graphene with Profs. Lin Lin, Mitchell Luskin, and Alexander Watson. Resulted in Refs. [12] and [20] above. Next position: William H. Kruskal Instructor at the University of Chicago.
- Raehyun Kim (Graduate Student from University of California, Berkeley; 2022–2025)
  - Co-supervised project on *ab initio* modeling for twisted bilayer graphene with Prof. Lin Lin. Resulted in [15]. Next position: Developer at Ansys, Inc.
- Zachary Carlini, Daniel Levy, Kate Samatovna, Henry Zou (Research in Industrial Projects for Students; IPAM REU Summer 2022)
  - Co-supervised, with researchers from IBM, an undergraduate research project on leveraging quantum teleportation to speed up quantum computation. Next positions:
    - Zachary Carlini: Graduate student at Yale
    - Daniel Levy: Graduate student at Princeton
    - Kate Samatovna: Master's student at Cambridge

  - Henry Zou: Position at IBM (follow up from summer project)

- Muthu Arivoli, Xinyu (Norah) Tan, and Chongbin Zheng (Undergraduates from Duke University; REU Summer 2019)
  - Managed undergraduate research project led by Alexander Watson on polarization in disordered materials. Next positions:
    - Xinyu Tan: Graduate student at MIT
    - Chongbin Zheng: Graduate student at Rice

Service

- Challenge Institute for Quantum Computation Organizing Committee (July 2023 – July 2026)
  - Organized weekly talks with the goal of bringing together the Berkeley quantum community and sparking discussions between different research groups.
  - Planned panel discussion at Annual Retreat 2025 on how to expand collaboration to other universities.
- Reviewer for: SIAM Multiscale Modeling & Simulation, SIAM Journal on Applied Mathematics, SIAM Journal on Mathematical Analysis, SIAM Journal on Scientific Computing, Journal of Mathematical Physics, Physical Review Letters, Physical Review B, European Physical Journal B, Japan Journal of Industrial and Applied Mathematics

Software

- **Programming Languages:** Python, Julia, Lisp, C/C++, MATLAB